

Development Methodology



4th June 2025

Note for the exam

Rules

This is an open-book exam: you are allowed to use handouts, notes, textbooks, and materials saved on your computer. However, if you use your computer for consultation, please ensure that airplane mode is activated.

Communication of any kind, whether among yourselves or with external parties, is strictly prohibited. Any violation of this rule will result in the invalidation of your exam.

Duration

1.5 hours.

How much should I write?

You should aim to write no more than 4 sides of paper. Use English and focus on including plenty of diagrams, charts, and visual aids, with minimal text. Prioritize clarity and conciseness to convey your ideas effectively. Use a black or blue pen; pencils or other colors are not allowed.

Pay attention

Write your name and student ID number at the top of the first page and clearly mark the sheet that represents the final version of your exam.

Any other questions?

Rise your hand if something is not clear, otherwise make (and write down) your assumptions.

Use case: Cloudbeds (<https://www.cloudbeds.com/>)

Introduction

Cloudbeds is an integrated platform designed to optimize the operations of hotels, hostels, and vacation rentals. It consolidates numerous services into a single user-friendly interface to improve efficiency. **All services** are required to be cloud-based rather than on-premise.

Objective

The objective of this case is to propose comprehensive architectural and organizational solutions to effectively implement Cloudbeds project.

Case Study Overview

Cloudbeds' offerings include:

- Comprehensive property management solutions for hotels, hostels, and vacation rentals
- Integrated reservation systems to streamline bookings
- Guest communication tools to enhance the customer experience
- Workflow optimization to increase operational efficiency
- Revenue management features to maximize profits

Architectural Solutions

System Integration

Integrating various subsystems into a unified system, the architecture should enable seamless data flow between these components and permit the implementation of the entire offering (refer to the “Case Study Overview” paragraph).

Considering that a customer may possess multiple hotels, it is anticipated that they will be able to efficiently manage all their properties through a **unified interface**. This approach ensures streamlined operations and better oversight of their entire portfolio, facilitating ease of access to necessary information and coordination across different locations.

Data Management

Importance of a Well-Defined Data Model: Establishing a robust and well-structured data model is critical to achieving effective customer management. It ensures that all systems operate with a unified view of customer-related activities.

Key Objectives:

- Enable 360-degree customer management by integrating data from all customer interactions and touchpoints.
- Ensure that all systems have access to clear, consistent, and comprehensive information to make informed decisions.
- Facilitate seamless communication and data sharing across services, ensuring operational efficiency and customer satisfaction.

Core Data to Include: Reservations, communication history, customer feedback, preferences, and transaction records.

Outcome: A unified data model empowers predictive analytics, personalized customer experiences, and optimized decision-making processes across all Cloudbeds services.

Security and Privacy

Implement encryption, secure access controls, and ensure the maintenance of data security and privacy to keep these aspects at the highest level. In the context of Cloudbeds, these measures are particularly crucial due to the sensitivity of data in the hospitality sector.

Encryption ensures that data is readable only by authorized parties, protecting it from unauthorized access during transmission and storage. Secure access controls involve managing who can view or modify the data, ensuring that only authorized personnel have access based on their roles and responsibilities.

Another fundamental aspect is room security, such as electronic door management. These systems allow monitoring and controlling access to rooms, guaranteeing a high level of security for guests. These steps collectively help safeguard personal guest information and business data from breaches or improper use.

Organizational Solutions

Project Management

It is essential to define the methodologies to be implemented in the construction of this system, as they play a pivotal role in aligning project execution with strategic goals and ensuring operational efficiency.

By selecting and clearly articulating the methodologies, the team can establish a structured framework that facilitates collaboration, streamlines processes, and enhances the overall effectiveness of the system's development.

Team Structure

To successfully implement the various services described, it is essential to form a multidisciplinary team equipped with the necessary skills and expertise.